

Ahmad Rahman

Computer Science Student | Web Development & Systems Programming

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PROFESSIONAL SUMMARY

Computer Science undergraduate with focused expertise in Web Development, Systems Programming, and Applied Deep Learning. Developed and spearheaded two major AI-focused initiatives: a Brain Tumor Detection and Explainable AI system (Scanvidence) using MRI data with Python, PyTorch/TensorFlow, ResNet50, EfficientNet, Vision Transformer (ViT), SHAP, LIME, and Grad-CAM for high-precision diagnostic classification and model interpretability; and GPT-Neo (Ahmad's Edition), a large-scale autoregressive language modeling project built with Python, TensorFlow, Mesh TensorFlow, and Transformer architecture, supporting model and data parallelism across GPUs/TPUs, custom dataset tokenization and training, text generation, and advanced techniques including local attention, linear attention, Mixture of Experts (MoE), and axial positional embeddings.

EDUCATION

Integral University — *Bachelor of Technology (B.Tech) in Computer Science and Engineering* Lucknow, India
Expected 2027

Awadh Educational Academy — *Class XII (Senior Secondary)* Lucknow, India
2023

RELEVANT COURSEWORK

Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems, Computer Networks

TECHNICAL SKILLS

Languages: Python, JavaScript, HTML, CSS, SQL, Java, C/C++

Frameworks & Libraries: React.js, Node.js, Express.js, FastAPI, PyTorch, TensorFlow, Mesh TensorFlow, Transformers, Tailwind CSS, Alpine.js, Bootstrap, SQLAlchemy

Developer Tools & Databases: Git, GitHub, VS Code, Jupyter, Google Colab, Postman, Docker, MongoDB, MySQL, SQLite, Power BI, Excel

Testing & Evaluation: Model Evaluation, Confusion Matrix, SHAP, LIME, Grad-CAM, API Testing, Functional Testing, Data Validation

EXPERIENCE

OctaNet — *Web Development Intern* Lucknow, India / Remote | 2024

- Engineered responsive web applications and full-stack features leveraging the MERN stack (MongoDB, Express.js, React.js, Node.js), JavaScript, HTML, CSS, and Bootstrap.
- Designed and integrated RESTful APIs to facilitate seamless data communication, user authentication, and data persistence between frontend clients and backend services.
- Utilized Git and GitHub for version control, collaborative code management, and agile feature delivery.

PROJECTS

EventPlan | *FastAPI, WebSockets, SQLAlchemy, SQLite, Tailwind, Alpine.js* [GitHub](#)

- Served as Frontend Developer for Integral University's Event Management System, building responsive, role-aware interfaces for student, organizer, and administrator workflows using Tailwind CSS and Alpine.js.
- Integrated REST APIs and WebSockets to support real-time event updates, live seat tracking, and registration notifications.
- Implemented multi-step registration, status tracking, QR-code ticketing, form validation, dynamic loading states, and robust error handling.

GPT-Neo (Ahmad's Edition) | *Python, TensorFlow, Mesh TensorFlow, Transformers* [GitHub](#)

- Engineered and trained a large-scale autoregressive language model utilizing Python, TensorFlow, and Mesh TensorFlow, enabling distributed model and data parallelism across GPU/TPU accelerators.
- Built end-to-end pipelines for custom dataset tokenization, sequence batching, model configuration, autoregressive training, and text generation.
- Explored and implemented advanced Transformer techniques including local attention, linear attention, Mixture of Experts (MoE), and axial positional embeddings to enhance computational scalability.

Scanvidence | *Python, PyTorch, TensorFlow, ViT, SHAP, LIME, Grad-CAM*

[GitHub](#)

- Engineered an automated deep learning diagnostic system for brain tumor detection and classification from MRI data using ResNet50, EfficientNet, and Vision Transformer (ViT) architectures.
- Implemented Explainable AI (XAI) algorithms including SHAP, LIME, and Grad-CAM to interpret model predictions and highlight crucial regions in MRI scans for clinical interpretability.
- Evaluated classification performance using confusion matrices and standard metrics, while exploring quantum computing methodologies as an extension for model optimization and feature selection.